

	1	2	3	4	5		
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch	Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch LED Panel Buffers File: led_panel_buffers.kicad_sch Pushbuttons File: pushbuttons.kicad_sch Mechanical File: mechanical.kicad_sch LED Panel Connectors File: led_panel_connectors.kicad_sch					A
B						B	
C						C	
D						D	
	1	2	3	4	5		

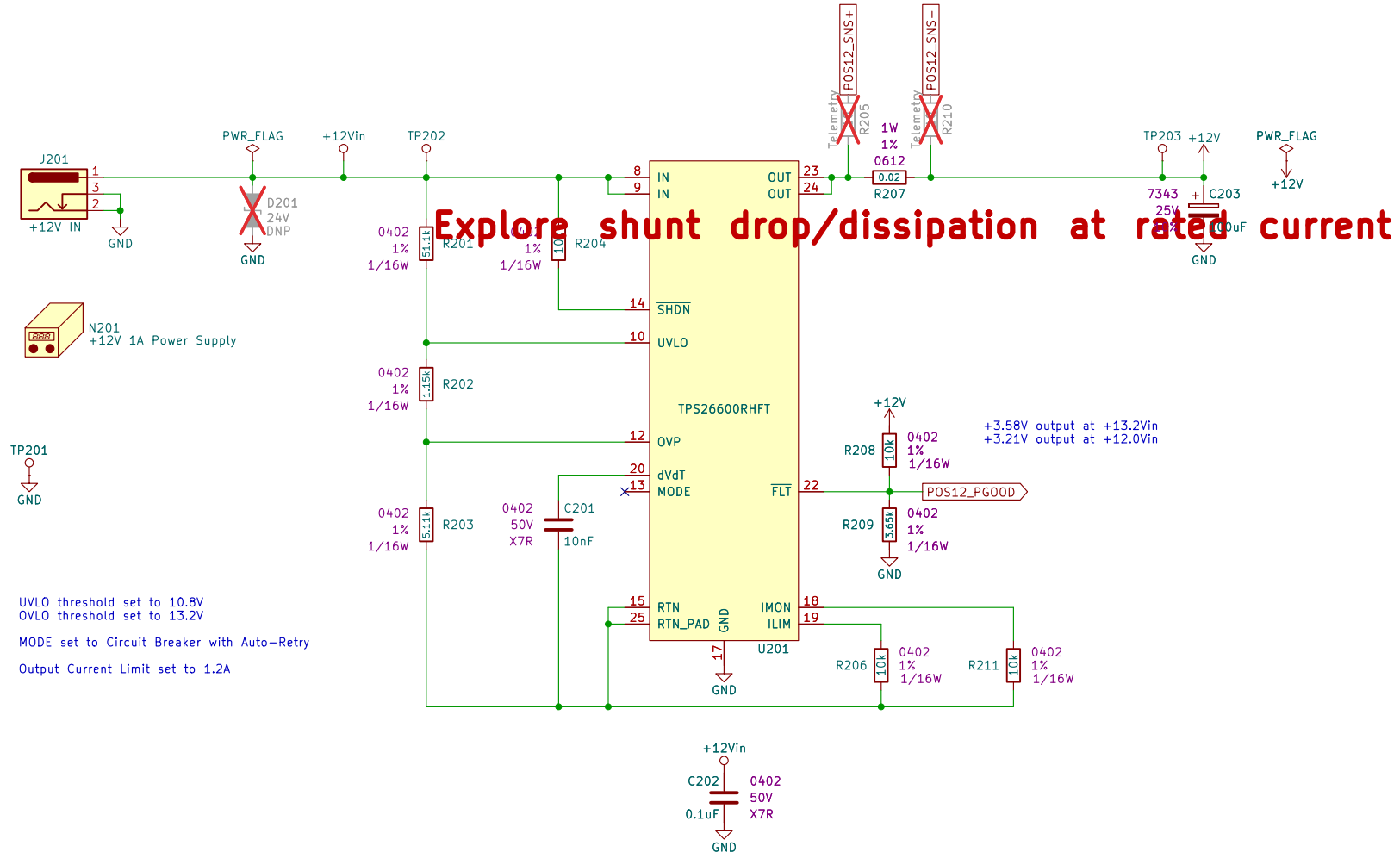
Sheet: /
File: Audio_Visualiser_DSP.kicad_sch

Title:

Size: ADate: 2023-12-25Rev:

KiCad E.D.A. 10.0.3Id: 1/21

1.2A is not enough, this device will most likely need to be changed



Load	Draw	Quantity	Total
LED Panel VLED 256 * 0.02A @ 3V = 15.36W / 12V = 1.28A / 80% efficiency	1.6A	2	3.2A
LED Panel POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	2	0.688A
Local POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	1	0.344A
Total		50.784W	4.232A

Sheet: /12V Power Input/
File: POS12_power_input.kicad_sch

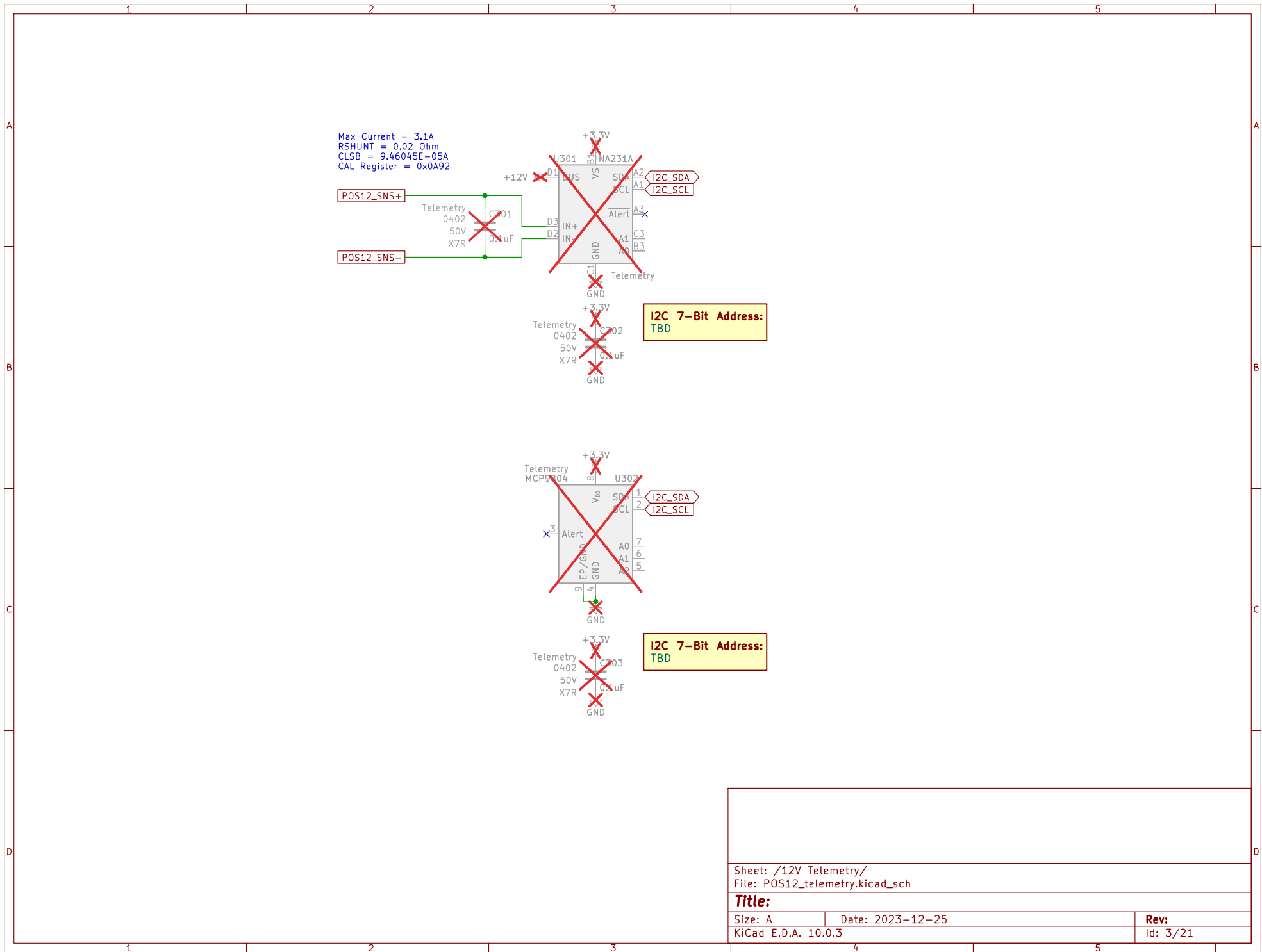
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Size: A Date: 2023-12-25

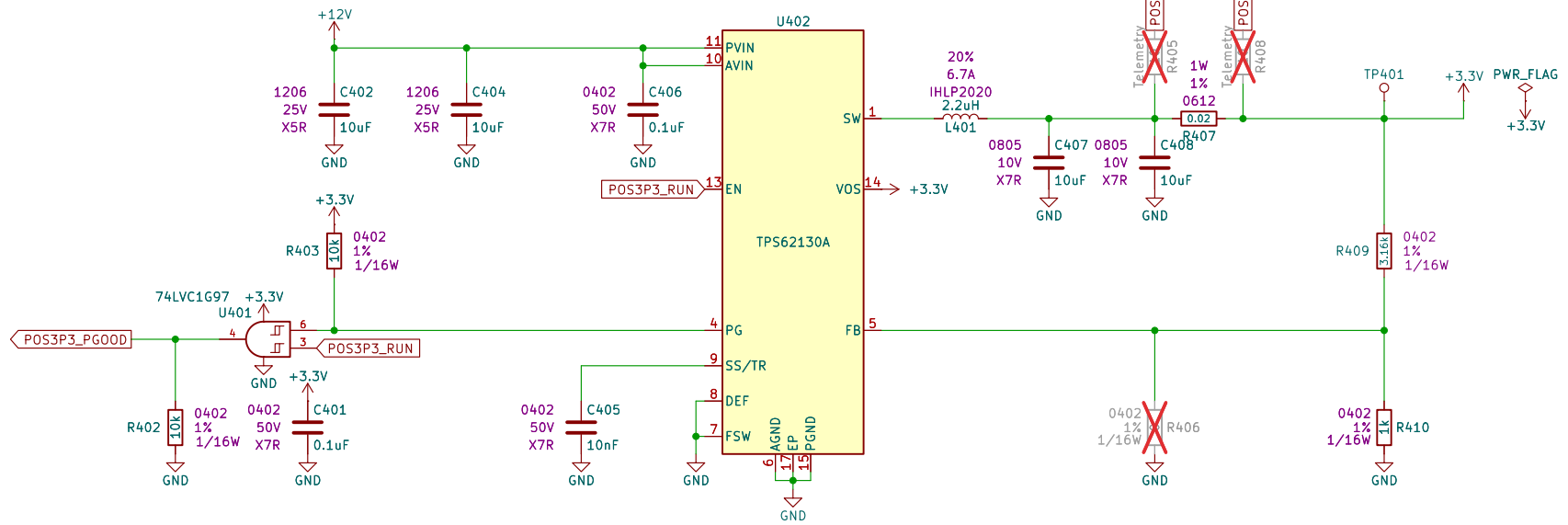
KiCad E.D.A. 10.0.3

Rev:

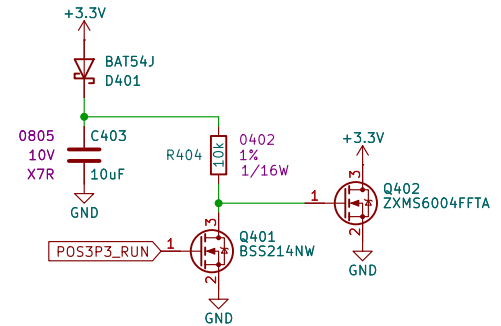
Id: 2/21



SHRINK



POS12_PG00D 0 POS3P3_RUN
R401



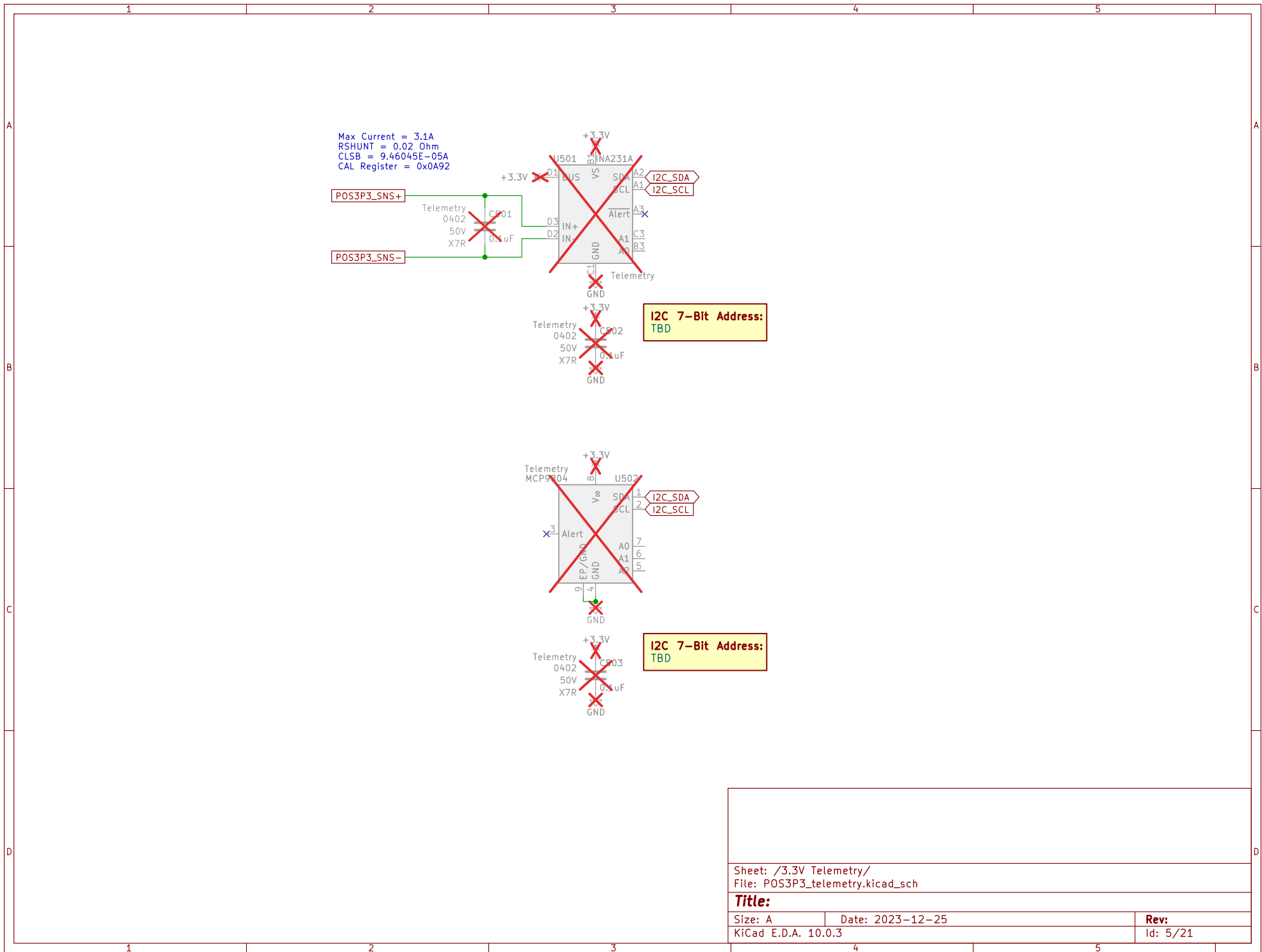
Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	4	4.4mA
MCP6024	5.4mA	2	10.8mA
Analog Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V Reference	0.2mA	2	0.4mA
PCA9536 GPIO Expander	0.29mA	1	0.29mA
I2C Buffers	8.3mA	3	24.9mA
Octal Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary Encoder	50mA	1	50mA
Module	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap Touch Sensor	0.729mA	1	0.729mA
Total		0.554W	168mA

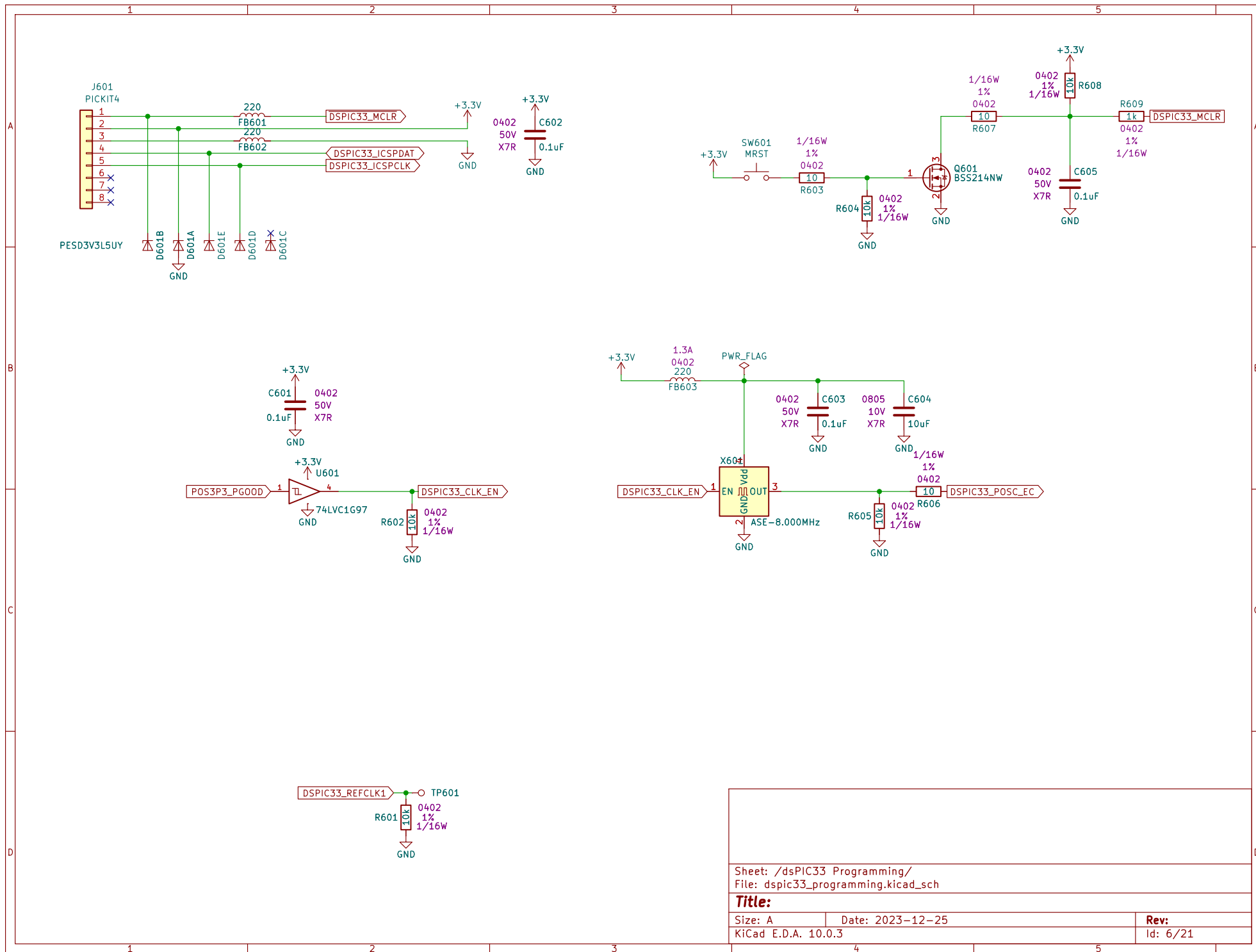
Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 4/21



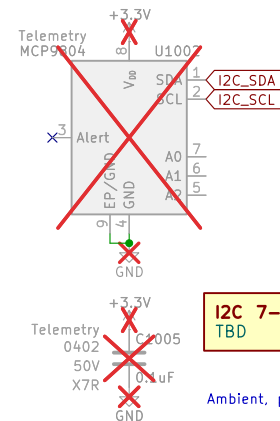
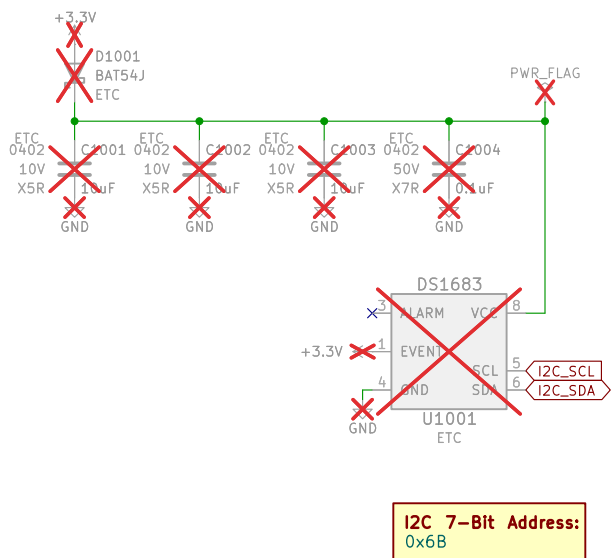


Sheet: /dsPIC33 Programming/
File: dspic33_programming.kicad_sch

Title:

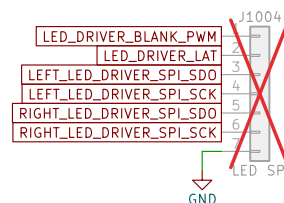
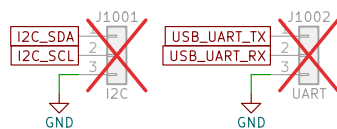
Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 6/21



Ambient, place away from power components

Test Points

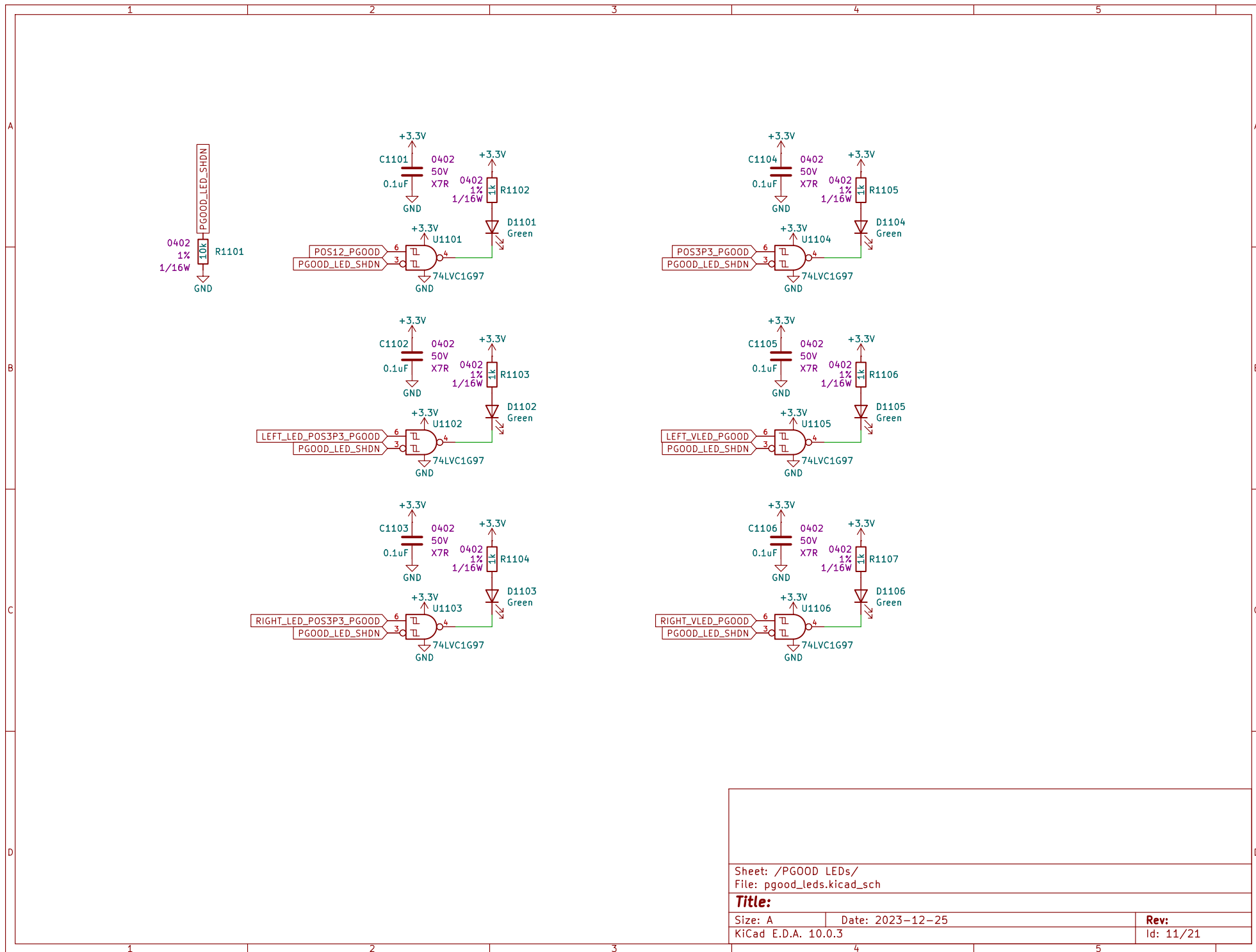


Sheet: /Platform ETC/
File: platform_etc.kicad_sch

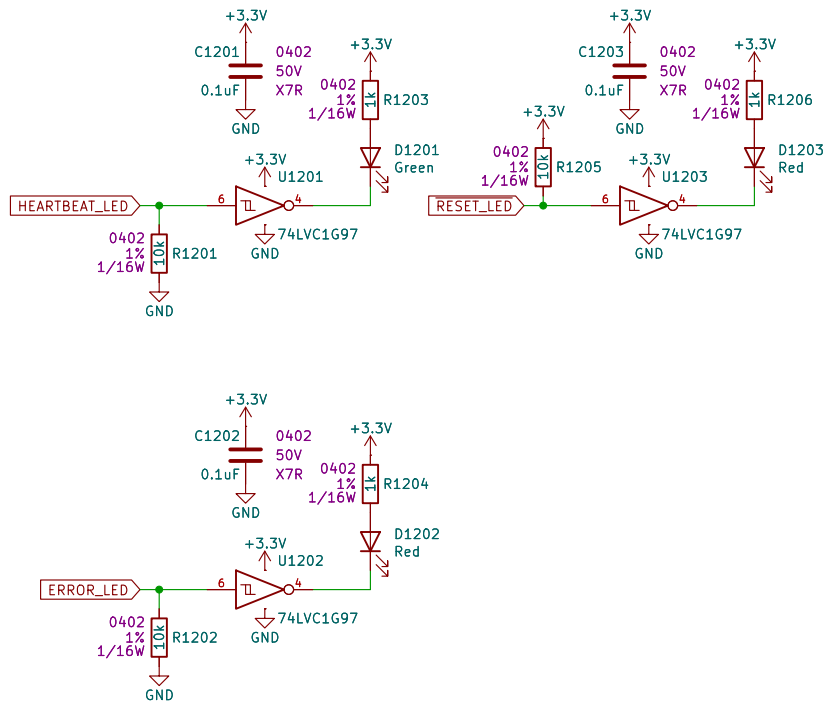
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Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 10/21



Sheet: /PGOOD LEDs/ File: pgood_leds.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 11/21

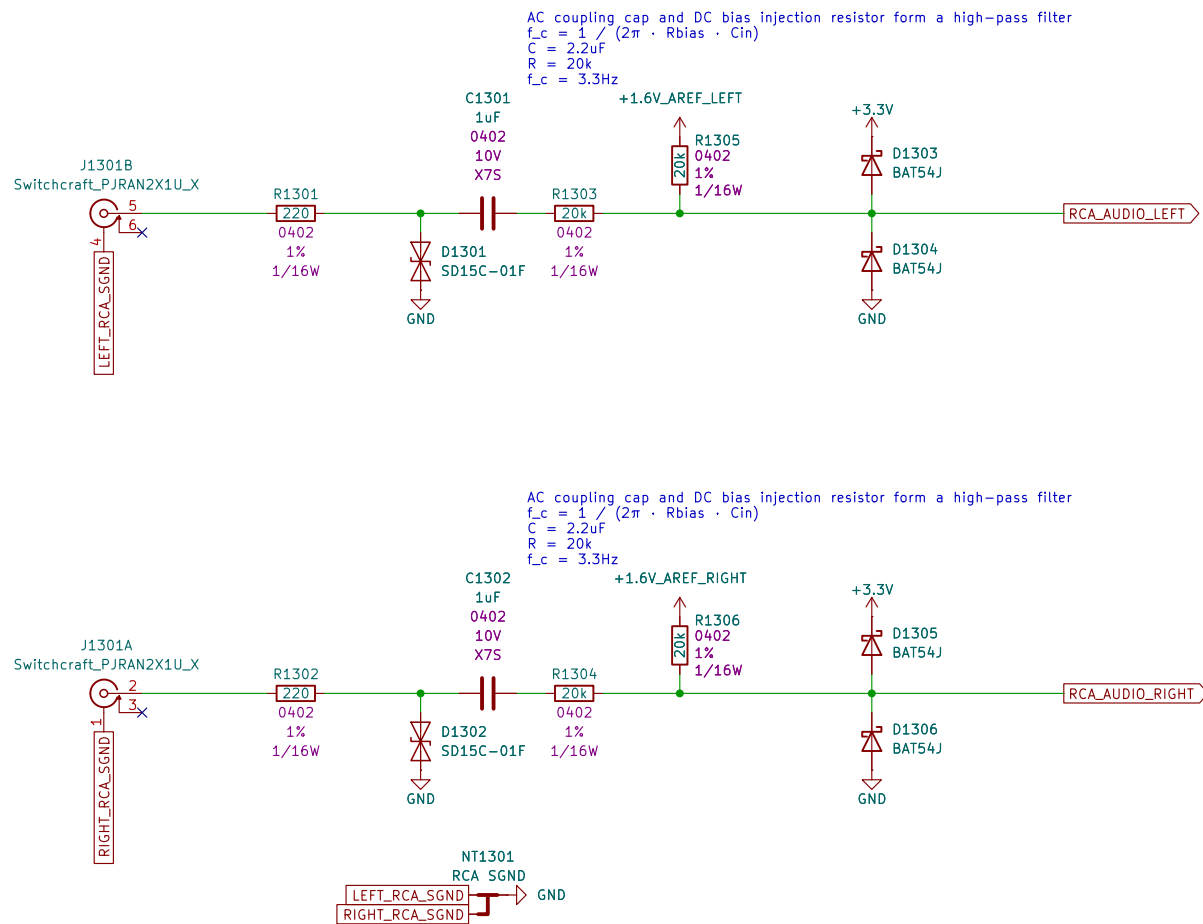


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File: status_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/21

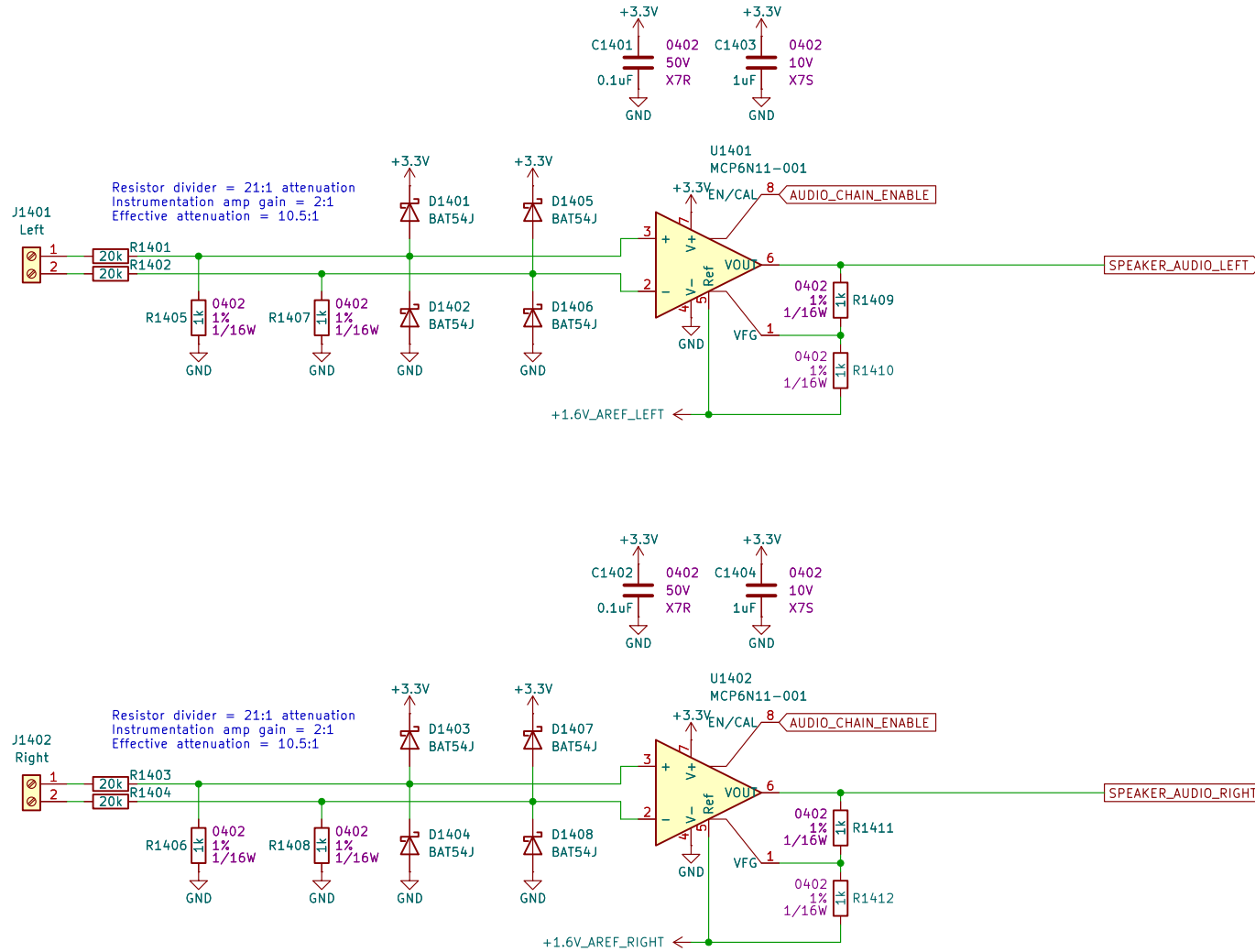


Sheet: /RCA Signal Conditioning/
 File: rca_signal_conditioning.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 13/21

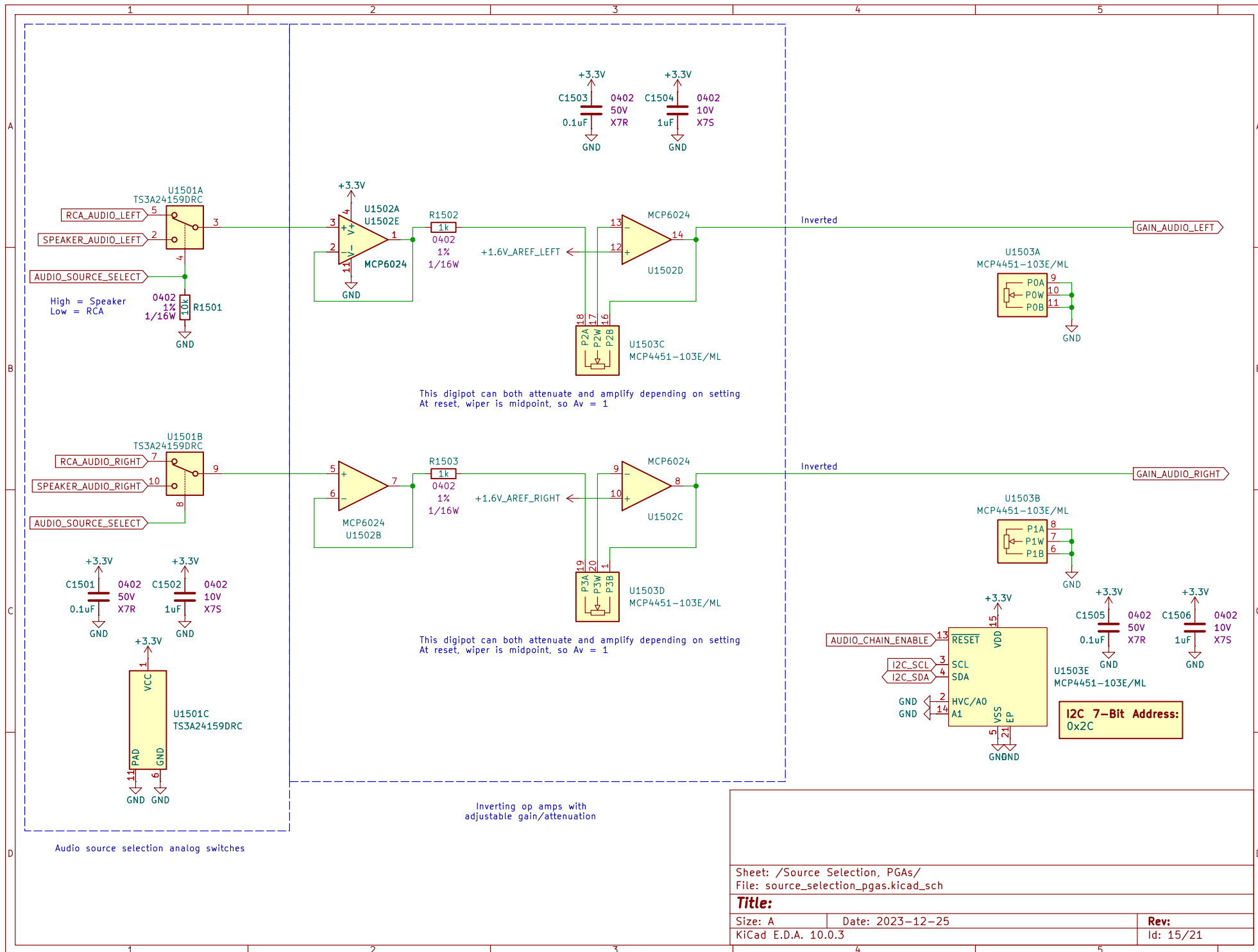


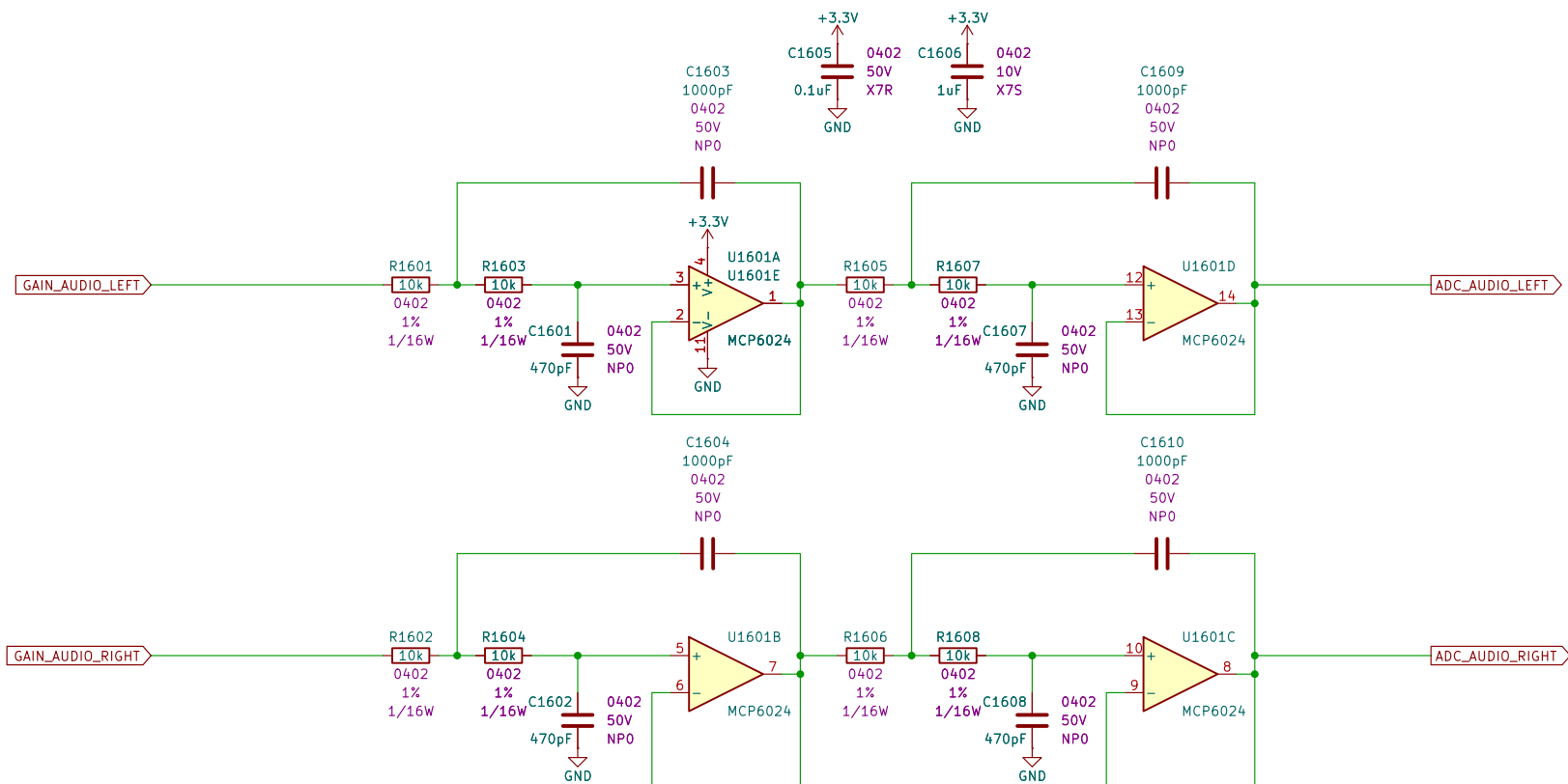
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File: speaker_signal_conditioning.kicad_sch

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Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 14/21



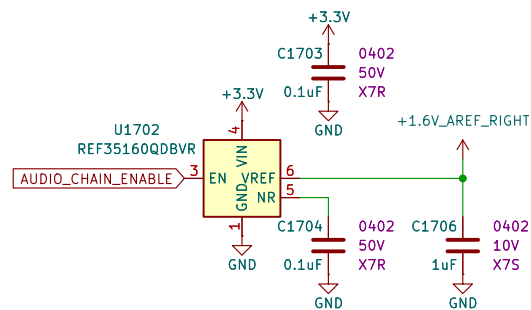
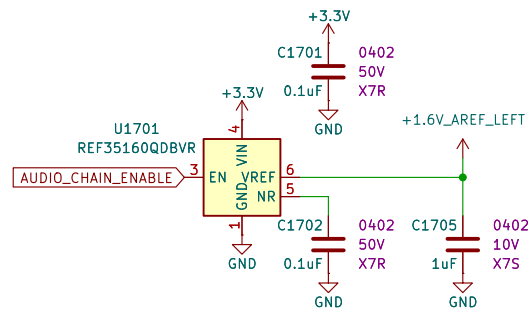


Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 16/21



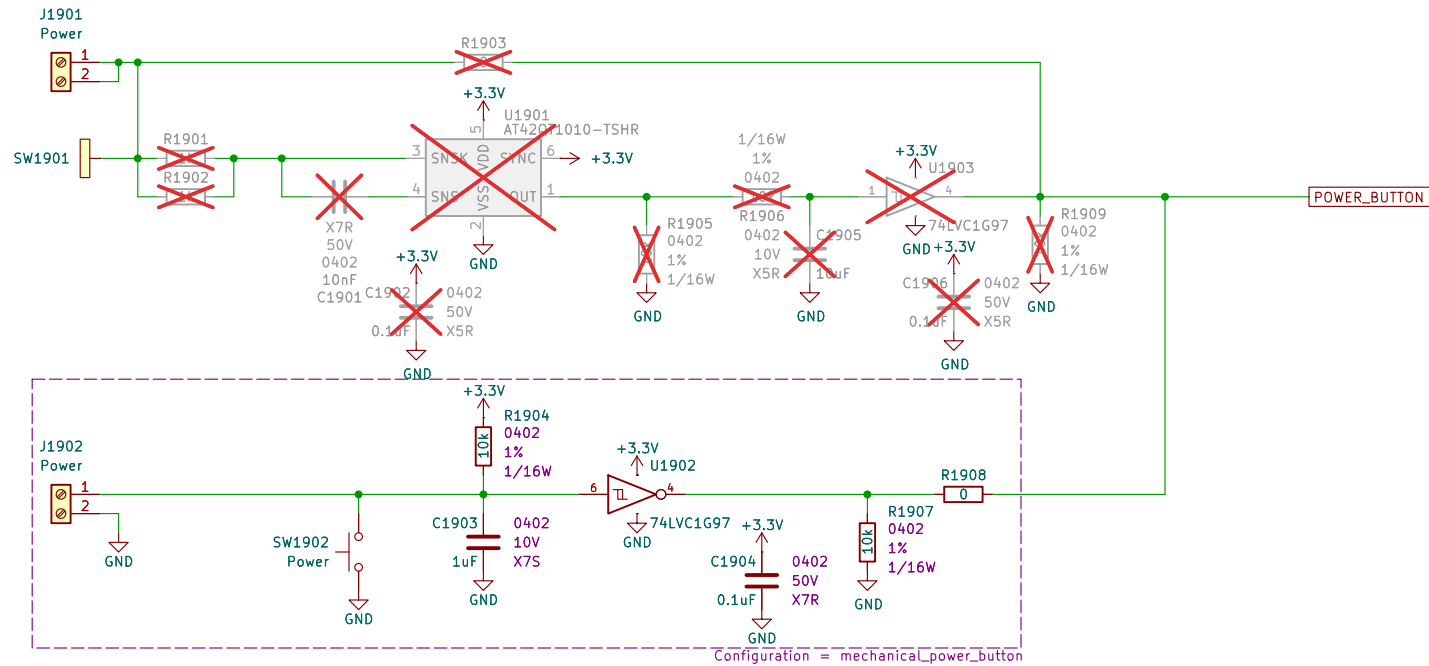
Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

Size: A Date: 2023-12-25

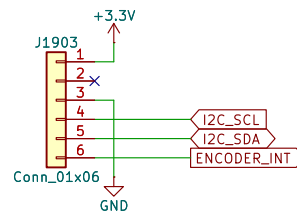
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Rev:
Id: 17/21



N1901
Adafruit Rotary Encoder Module

I2C 7-Bit Address:
0x36



Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

Title:

Size: A Date:

KiCad E.D.A. 10.0.3

Rev:

Id: 19/21

